

Coach Kai Chat Fix - December 28, 2025

Problem

Coach Kai text chat was completely broken, returning “Sorry, I had trouble with that. Please try again!” for ALL messages.

Root Cause Analysis

The API routes were looking for an environment variable called `ABACUSAI_API_KEY`, but the system has `ABACUS_API_KEY` configured as the standard Abacus.AI authentication variable.

Investigation Steps

1.  Examined frontend component (`components/coach/simple-coach-kai.tsx`)
 - Found it calls `/api/ai-coach/chat` endpoint
2.  Analyzed API route (`app/api/ai-coach/chat/route.ts`)
 - Found API checks for `process.env.ABACUSAI_API_KEY` at the beginning
 - Returns 503 error if not found
3.  Checked environment variables
 - Confirmed `.env` file does NOT have `ABACUSAI_API_KEY`
 - Confirmed system HAS `ABACUS_API_KEY` environment variable
4.  Identified the mismatch
 - API expects: `ABACUSAI_API_KEY`
 - System provides: `ABACUS_API_KEY`

Solution

Updated ALL API routes to use the correct environment variable name `ABACUS_API_KEY`.

Files Modified

1. `app/api/ai-coach/chat/route.ts` - Main chat endpoint
2. `app/api/coach-kai/enhanced/route.ts` - Enhanced chat endpoint
3. `app/api/training/generate-program/route.ts` - Training program generation
4. `app/api/ai-coach/recommendations/route.ts` - AI recommendations
5. `app/api/ai-coach/daily-coaching/route.ts` - Daily coaching
6. `app/api/ai-coach/insights/route.ts` - AI insights
7. `app/api/tts/openai/route.ts` - Text-to-speech
8. `lib/ai/abacus-client.ts` - Abacus AI client library

Testing

-  Build succeeded without errors
-  Ready for deployment to production

Impact

This fix resolves the complete failure of Coach Kai chat functionality. Users will now be able to:

- Ask questions and receive AI responses
- Get coaching advice
- Use quick topic buttons
- Interact with all Coach Kai features

Next Steps

1. Deploy to production (Vercel)
2. Verify chat works on live site
3. Monitor for any errors in production logs